

Active record Associations

Original Author: Dr. Diego Martinez, TA, 2021-2022 AY

https://guides.rubyonrails.org/association_basics.html

Goal:

Create a new CRUD in your Book Collection app that interacts with a 'join table' (i.e., associative entity) which results from two entities having a many to many relationships.

Steps (selected):

1. Create a new entity using scaffold, the new entity is User with a single attribute username:string
2. Modify the attributes to represent many to many users to books
3. Create a new entity (i.e., associative entity) that contains user_id:integer and book_id:integer (hint: scaffold is a great tool)
4. Modify the new view (titled "User Books") on the new entity to show only the available users and books.
5. Add links to the book and users indexes.
6. Modify the routes file so the root route goes to the new view mentioned in the previous item.
7. Push to the Github classroom, using link below:
[Book Collection 3-4](#)

Hints :

- Create a new entity using scaffold, the new entity is User with a single attribute username:string

rails g scaffold User username:string

rails g scaffold UserBook user_id:integer book_id:integer

- Modify the attributes to represent many to many users to books

```
1 class Book < ApplicationRecord
2   has_many :users, through: :user_books
3   has_many :user_books
4
5   validates :title, :author, :price, :published_date, presence: true
6 end
7
```

```
class UserBook < ApplicationRecord
  belongs_to :user
  belongs_to :book
end
```

```
1 class User < ApplicationRecord
2   has_many :books, through: :user_books
3   has_many :user_books
4
5 end
6
```

- Create a new entity that contains user_id:integer and book_id:integer (hint: scaffold is a great tool) (**done above**)

- Modify the new view on the new entity to show only the available users and books.
- Add links to the book and users indexes.

```

1 <p id="notice"><%= notice %></p>
2
3 <h1>User Books</h1>
4
5 <table>
6   <thead>
7     <tr>
8       <th>Username</th>
9       <th>Book Title</th>
10      <th colspan="3"></th>
11    </tr>
12  </thead>
13
14  <tbody>
15    <%= @user_books.each do |user_book| %>
16      <tr>
17        <td><%= User.where(id: user_book.user_id).last.username %></td>
18        <td><%= Book.where(id: user_book.book_id).last.title %></td>
19        <td><%= link_to 'Show', user_book %></td>
20        <td><%= link_to 'Edit', edit_user_book_path(user_book) %></td>
21        <td><%= link_to 'Destroy', user_book, method: :delete, data: { confirm: 'Are you sure?' } %></td>
22      </tr>
23    <%= end %>
24  </tbody>
25 </table>
26
27 <br>
28
29 <%= link_to 'New User Book', new_user_book_path %>
30
31 <br>
32
33 <%= link_to 'Book index', books_path%>
34
35 <br>
36
37 <%= link_to 'User index', users_path%>
38

```

- Modify the routes file so the root route goes to the new view

```

1 Rails.application.routes.draw do
2   resources :user_books
3   resources :users
4   root 'user_books#index'
5
6
7   resources :books do
8     member do
9       get :delete
10    end
11  end
12  # For details on the DSL available within this file, see https://guides.rubyonrails.org/routing.html
13 end
14

```

You need to add

```
<%= form.select :user_id, User.order(:username).map { |user| [user.username, user.id] },  
include_blank: true, id: "id_of_the_user_id_select_box" %>
```

(shown in the screenshot below)

In app/views/user_books/_form.html.erb and also you need to edit one more form field in the same file identify and do. (you need to do this so that localhost:3000/user_books/new has dropdown list of all available users and books)

```
1 <%= form_with(model: user_book) do |form| %>  
2   <% if user_book.errors.any? %>  
3     <div id="error_explanation">  
4       <h2><%= pluralize(user_book.errors.count, "error") %> prohibited this user_book from being saved:</h2>  
5       <ul>  
6         <% user_book.errors.each do |error| %>  
7           <li><%= error.full_message %></li>  
8         <% end %>  
9       </ul>  
10    </div>  
11  <% end %>  
12  
13  <div class="field">  
14    <%= form.label :user_id %>  
15    <%= form.select :user_id, User.order(:username).map { |user| [user.username, user.id] }, include_blank: true, id: '  
16  </div>  
17  
18  <div class="field">  
19    <%= form.label :book_id %>  
20    <%= form.select :book_id, Book.order(:title).map { |book| [book.title, book.id] }, include_blank: true, id: "id_of_  
21  </div>  
22  
23  <div class="actions">  
24    <%= form.submit %>  
25  </div>  
26  
27 <% end %>  
28
```

```
<%= form.select :user_id, User.order(:username).map { |user| [user.username,  
user.id] }, include_blank: true, id: "id_of_the_user_id_select_box" %>
```

```
<%= form.select :book_id, Book.order(:title).map { |book| [book.title, book.id] },  
include_blank: true, id: "id_of_the_book_id_select_box" %>
```

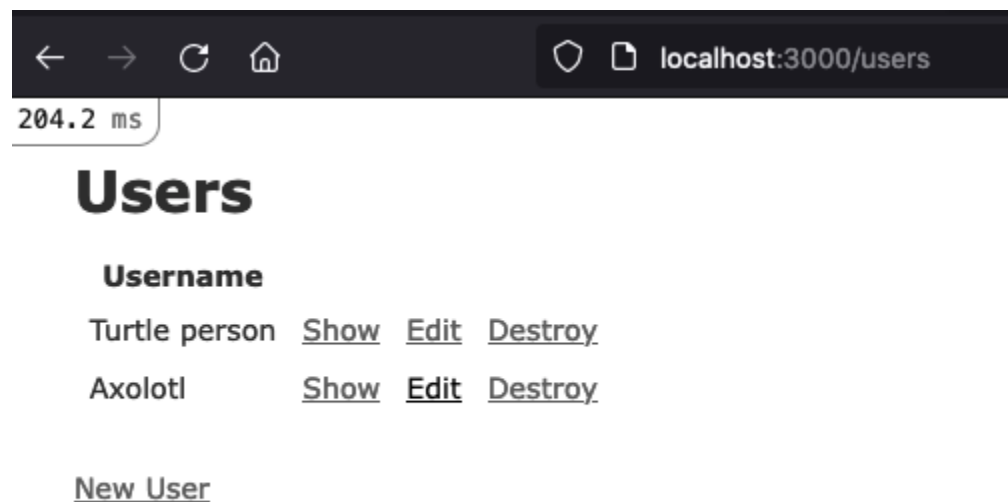
Submission:

- Provide the screenshot of the user view (1)
- Provide the screenshot of creating a new user_book instance (2)
- Provide the screenshot of the new view (titled "User Books") which shows only the available users and books, with links to book index, user index and new user_book. (3)
- Provide the screenshot of the routes file, showing user_book index as the root route
- Provide screenshots of merging changes into the proper Github branches
- Provide screenshots of commit history
- Submit a screenshot of an analysis of your code against the Canvas rubric using the LLM of your choice. This should communicate improvement suggestions to comply with the given rubric.

Save your report as FirstName_LastName_Lab-BookCollection4.pdf and submit it to Canvas.

Final views should look like this :

1.)



2.)

← → ↻ 🏠 144.5 ms localhost:3000/user_books/new

New User Book

User
Axolotl ▾

Book
Amazing turtles ▾

Create User book

[Back](#)

3.)

← → ↻ 🏠 395.7 ms localhost:3000

User Books

Username	Book Title	
Turtle person	Amazing turtles	Show Edit Destroy
Axolotl	The secret of life	Show Edit Destroy
Axolotl	Amazing turtles	Show Edit Destroy

[New User Book](#)
[Book index](#)
[User index](#)

Grading

Will be graded using the following sections of the Sprint Rubric:

- **Data Design**

If you are facing the below error, you can try running this command :

Webpacker::Manifest::MissingEntryError in Books#index

Showing /csce431/book-collection-part-2-capito2000/app/views/layouts/application.html.erb where line #10 raised:

Webpacker can't find application.js in /csce431/book-collection-part-2-capito2000/public/packs/manifest.json. Possible causes:

1. You want to set webpacker.yml value of compile to true for your environment unless you are using the 'webpack -w' or the webpack-dev-server.
2. webpack has not yet re-run to reflect updates.
3. You have misconfigured Webpacker's config/webpacker.yml file.
4. Your webpack configuration is not creating a manifest.

Your manifest contains:

```
{  
}
```

Extracted source (around line #10):

```
8  
9 <%= stylesheet_link_tag 'application', media: 'all', 'data-turbolinks-track': 'reload' %>  
10 <%= javascript_pack_tag 'application', 'data-turbolinks-track': 'reload' %>  
11 </head>  
12  
13 <body>
```

Rails.root: /csce431/book-collection-part-2-capito2000

[Application Trace](#) | [Framework Trace](#) | [Full Trace](#)

[app/views/layouts/application.html.erb:10](#)

Request

Parameters:

None

[Toggle session dump](#)

bundle exec rake webpacker:install